Lab N: Title of the lab

Course Name

Student: Your Name — YOUR-ID

1. Introduction

What the lab asks for, in two or three sentences. The companion notebook is `Lab1.ipynb`. Every number in this report comes from its saved output.

Word	Meaning here
step size	how far each update moves

2. Problem and data

The problem in math, and the data (size, split, source).

Key formulae

$$\min_x \; f(x) = \tfrac{1}{2}\|Ax - b\|_2^2 + \lambda\|x\|_2^2$$

3. Step 1: Method name

Idea

One paragraph in plain words.

Math and update rule

$$x^{(k+1)} = x^{(k)} - t\,\nabla f(x^{(k)})$$

Method

```
for k in range(max_iter):
    x = x - t * grad(x)
```

Listing 1: Gradient descent loop (logging left out).

Results

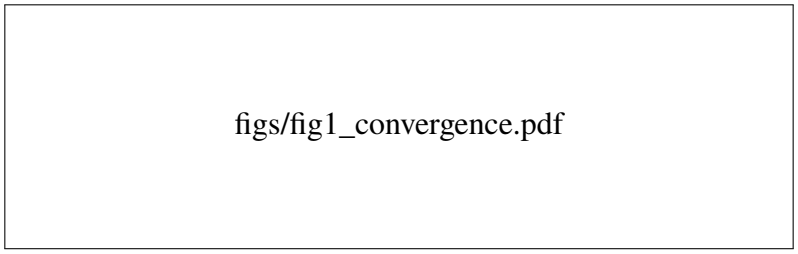

Figure 1: What the figure shows, and the one thing to notice.

Does it reach the Step-1 optimum? Say yes or no, with the number.

4. Comparing all methods

Method	Iterations	Final error	Time (s)
GD	120	1.2×10^{-6}	0.03

Trade-offs. Which method to pick and when.

5. Conclusion

Three or four sentences: what worked, what didn't, and why.